

Table 12.5: Index Value in Mean Standardization Method

District	Percentage of other than agricultural labour to all labour	Percentage of net irrigated area	Per capita value of agricultural output	Per capita consumption expenditure	Female Literacy rate	women work force participation rate	Percentage of household getting safe drinking water	Percentage of villages having access to paved road	Percentage of villages having access to primary health centre	Average casual wage rate	Composite Index
Malkangiri	1.237	1.332	1.899	0.808	0.561	1.079	1.379	0.614	0.775	1.003	1.069
Rayagada	0.847	0.988	0.697	0.808	0.561	1.079	1.312	0.995	0.630	1.030	0.895
Sundargarh	1.034	0.816	0.782	1.126	1.340	0.934	0.959	1.112	1.357	0.891	1.035
Nabarangapur	0.780	0.258	1.222	0.808	0.561	1.063	1.346	1.316	1.357	1.030	0.974
Kandhamal	1.051	0.602	0.597	0.808	1.028	1.095	0.538	0.614	0.921	1.003	0.826
Koraput	0.932	1.332	0.890	0.808	0.499	1.079	1.127	0.644	1.066	1.114	0.949
Mayurbhanj	1.017	1.289	0.833	1.126	1.091	0.999	0.740	1.229	1.018	0.863	1.020
Gajapati	0.881	1.074	0.770	1.331	0.748	1.240	0.723	0.848	0.969	0.947	0.953
Sambalpur	1.051	1.461	1.565	1.126	1.558	0.999	0.942	0.907	0.630	1.086	1.132
Kendujhar	1.017	0.988	0.782	1.126	1.371	0.741	0.875	1.346	0.727	0.947	0.992
Jharsuguda	1.153	0.859	0.964	1.126	1.683	0.693	1.060	1.375	1.551	1.086	1.155

12.5.4 Range Equalization Method

Range Equalization (RE) method otherwise known as max-min approach is adopted by UNDP in computation of Human Development Index. Under this approach, an index is constructed for each variable by applying Range Equalization formula derived by UNDP. This is worked out by subtracting an indicator's minimum value from each observation and then dividing it by its range.

$$\text{RE Index} = \frac{X_i - \text{Min } X}{\text{Max } X - \text{Min } X}$$

where X_i Value of the variable, min X- Minimum value of X in the scaling , max X- Maximum value of X in the scaling. The RE index is also a normalization technique. Without it, a composite index can be biased towards an indicator with very high range. For example the per capita value of agricultural output which is measured in rupees ranges from Rs. 410/- to Rs. 1304/-. On the other hand, the variable proportion of female literacy rate is measured in percentage. Different variables measured in different units sometimes give upward or downward biases. To ensure the index value biasfree, we convert all the variables into equal scaling from 0 to 1.

In RE method, as a first step we undertake scaling exercise. In undertaking the scaling procedure, desirable norms are followed for each indicator. In some cases, scaling of indicators is self-selected, while in others element of value judgment is involved. This scaling exercise is called goal post where we identify the maximum and minimum goal. The goalpost basically visualizes the extent of minimum value and maximum value in the future time period (say next 5 years). The scaling norm that we have adopted is given in Table 12.6. In the table the third and fourth column shows the maximum and minimum district value of selected 10 indicators. The first and second column is the maximum and minimum goal post.

Table 12.6: Construction of Food Security Radar

Variable	Description of Variables	Goalposts		District Value	
		Minimum	Maximum	District Minimum	District Maximum
Oth_agl	Percentage of other than agricultural labour to all labour	30	85	46	73
Irr	Percentage of net irrigated area to net cropped area	2	55	6	34
pcvao	Per capita value of agricultural output	200	2500	410	1304
Mpcce_ia	Inequality adjusted per capita consumption expenditure	150	450	201	331
Lit_f	Female Literacy (adult) rate	10	70	16	54
Wfpr_f	women work force participation rate	30	85	43	77
Hhsw	Percentage of household having access to safe drinking water	20	90	32	82
Paved_r	Percentage of villages having access to paved road	10	60	21	47
v_phcs	Percentage of villages having access to Primary health centre	10	50	13	32
Wage_c	average Casual wage rate	25	70	31	40

Based on the goalpost, we normalize the value of the variable for all the districts and all the indicators. In our example, the values have been presented in Table 12.7.

Table 12.7: Index Value in Range Equalization Method**Construction of Composite
Index in Social Sciences**

District	Percentage of other than agricultural labour to all labour	Percentage of net irrigated land to net sown area	Per capita value of agricultural output	Per capita consumption expenditure	Female Literacy rate	women work force participation rate	Percentage of households getting safe drinking water	Percentage of villages having access to paved road	Percentage of villages having access to primary health centre	Average casual wage rate	Composite Index
Malkangiri	0.782	0.547	0.480	0.170	0.133	0.673	0.886	0.220	0.150	0.244	0.429
Rayagada	0.364	0.396	0.121	0.170	0.133	0.673	0.829	0.480	0.075	0.267	0.351
Sundargarh	0.564	0.321	0.147	0.433	0.550	0.509	0.529	0.560	0.450	0.156	0.422
Nabarangapur	0.291	0.075	0.278	0.170	0.133	0.655	0.857	0.700	0.450	0.267	0.388
Kandhamal	0.582	0.226	0.091	0.170	0.383	0.691	0.171	0.220	0.225	0.244	0.300
Koraput	0.455	0.547	0.179	0.170	0.100	0.673	0.671	0.240	0.300	0.333	0.367
Mayurbhanj	0.545	0.528	0.162	0.433	0.417	0.582	0.343	0.640	0.275	0.133	0.406
Gajapati	0.400	0.434	0.143	0.603	0.233	0.855	0.329	0.380	0.250	0.200	0.383
Sambalpur	0.582	0.604	0.380	0.433	0.667	0.582	0.514	0.420	0.075	0.311	0.457
Kendujhar	0.545	0.396	0.147	0.433	0.567	0.291	0.457	0.720	0.125	0.200	0.388
Jharsuguda	0.691	0.340	0.201	0.433	0.733	0.236	0.614	0.740	0.550	0.311	0.485

After calculating the index of each variable, we have averaged them to give each of the five dimensions of food security. The composite food security index is again derived by averaging the five dimensions.

The normalized value for the variable percentage of other than agricultural labour to total worker' for Malkangiri district is found out in the following manner.

$$\frac{(\text{Actual value (73)} - \text{Minimum goalpost (30)})}{(\text{Maximum goalpost (85)} - \text{Minimum goalpost (30)})} = 0.782$$

Likewise we have converted and worked out the normalized value of all the variables and for all the districts shown in Table 12.7.

Comparing MS Method and RE Method

The composite index value worked out by both the RE and MS methods can be compared and analyzed (Table 12.8). We can also examine correlation between two indices. Here the correlation between final index value worked out by both the methods is very high (0.991).

Table 12.8: Comparison between RE and MS Methods

District	Composite Index Value by RE method	Composite Index Value by MS method
Jharsuguda	1.155	0.485
Sambalpur	1.132	0.457
Malkangiri	1.069	0.429
Sundargarh	1.035	0.422
Mayurbhanj	1.020	0.406
Kendujhar	0.992	0.388
Nabarangapur	0.974	0.388
Gajapati	0.953	0.383
Koraput	0.949	0.367
Rayagada	0.895	0.351
Kandhamal	0.826	0.300

Note: Correlation of index value is 0.991

The high degree of the correlation between the values of composite index computed by applying both methods separately indicate the high degree of homogeneity between the two approaches. However, the Range equalization method is preferred because it accounts for wider variations and strong correlations to the PCA composite.

One important point should be kept in mind that the index value as discussed above are based on equal weight. We can also give weights to variables/components discussed in section 12.5.

Check Your Progress 2

- 1) State the procedure to workout Composite Index by way of Simple Ranking Method.

.....

.....

.....

- 2) How will you compute the final value of Composite Index by Indices Method?

.....

.....

.....

- 3) What is the Distinction between Range Equalization Method and Mean Standardisation Method?

.....

.....

.....

.....

12.6 PRINCIPAL COMPONENT ANALYSIS (PCA)

PCA is one of the important composite index method used to analyse the various problems in social sciences. PCA is a mathematical technique that transfers a number of correlated variables into smaller number of uncorrelated variables called the principal components. For our analysis we have an array of variables which may have high correlation with each other. In case where there is high relation between variables, the PCA is a suitable technique where the correlation between different components is low.

Categorical data are not suitable for PCA analysis because in such cases categories converted into quantitative scale has no meaning. To avoid this type of problems, we should recode these variables into binary variables. Let us take an example of social category of household that are Scheduled caste (1), Scheduled tribe (2), Other backward caste (3) and General (4). In PCA analysis such type of variables have no meaning. Such type of categorical variables can be converted into a binary variable. For example if we want to study the dalit status, variable can be converted to a binary variable as dalit (code 2) and non-dalit (code 1, 3, and 4). Likewise, if we want to study the impact of reservation for other backward castes we can categorize as OBC (code 3) and other than OBC (code 1, 2 and 4).

The PCA is a data reduction technique. Under this technique, the original data set is transformed into a new set of uncorrelated variables called principal components. PCA reduces the number of variables in a data set to smaller number of dimension/s. In a series of variables if a_{mn} is the weight of m^{th} principal component and n^{th} variable then the matrix will be:

$$\begin{bmatrix} PC_1 \\ PC_2 \\ PC_3 \\ \vdots \\ PC_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ \vdots \\ X_n \end{bmatrix}$$

The matrix can be transformed into equation for principal component i.e.

$$PC_1 = a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n$$

$$PC_2 = a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n$$

.....

$$PC_n = a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n$$

The components are ordered so that the first component (PC_1) explains the largest possible amount of variation in the original data, subject to the constraint that the sum of the squared weights of the vectors ($a_{11}^2 + a_{12}^2 + \dots + a_{1n}^2$) is equal to one.

In the output the eigen value is one of the important determinant of PCA. The eigen value is a number that tell you how much variance there is in the data set. In other words the eigen value is a number telling us how spread out the data is on the line. The eigen vector with the highest eigen value is therefore the principal component.

As the sum of the Eigen values equals the number of variables in the initial data set, the proportion of the total variation in the original data set accounted by each principal component is given by λ_i/n . The second component (PC_2) is completely uncorrelated with the first component, and explains additional but less variation than the first component, subject to the same constraint. Each subsequent component captures additional dimension in the data and adds smaller and smaller proportion of variation in the original variables.

Before conducting PCA, we should run correlation between variables. If the correlation between two variables is very high, we may remove one of those variables. The reason being that the two variables seem to measure same thing. The higher the degree of correlation among variables, the lower will be the number of components required to capture common information. Sometimes two variables may be combined in some ways (taking average). The PCA can be applied either to the original values of variables or to the normalized values of the variables.

In general, normalization can be done by three methods, i.e (i) by deviation of the variables from their respective means (i.e.); (ii) by dividing the actual values by their respective means; (iii) and deviation of value of a variable from the mean which is then divided by standard deviation {i.e. $(\cdot)/\sigma$ }. We are applying here the second method.

Let us try to apply and analyse PCA by using the database given in Table 12.1. We are applying here the second method for normalization that is found out in Table 12.5 column 2 to 10.

Table 12.9: KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.394
Bartlett's Test of Sphericity	Approx. Chi-Square	47.933
	Df	45
	Sig.	.355

When the normalized database is prepared, you have to apply Kaiser-Meyer-Olkin (KMO) technique and Measure of Sampling Adequacy and Bartlett's Test of Sphericity (BTS). In KMO technique, the value varies between 0 and 1 and the value closer to 1 is better. However, according to some statisticians, the minimum value should be 0.6. The BTS tests the null hypothesis that the correlation matrix is an identity matrix. Let us remember that identity matrix is a matrix in which the main diagonal elements are 1 and off-diagonal elements are 0.

The above two tests provide the minimum standard before conducting any PCA.

12.6.1 Conducting PCA and Analyzing Results

For the purpose of Principal Component analysis and interpreting its results, SPSS is user friendly software. However, you can try to run the same in STATA also. Let us take our previous example of Table 12.5. In the table, there are 10 standardised variables and we can run our PCA model. Table 12.10 give the first output called 'communalities'

Table 12.10: (Output 1) Communalities: Extraction by Principal Component Analysis.

Variable name	Initial	Extraction
% of other than agricultural labour to all labour	1.000	.721
% of net irrigated area to net sown area	1.000	.748
Per capita value agricultural output	1.000	.710
Monthly per capita consumption expenditure	1.000	.702
Female Literacy rate (adult)	1.000	.876
women work force participation rate	1.000	.858
% of household access to safe drinking water	1.000	.789
% of villages having access to paved road	1.000	.773
% of villages having access to primary health centres	1.000	.572
Average Casual wage rate	1.000	.491

Communalities represent the percentage of variance explained by the extracted components. This explains the proportion of each variable's variation explained by PCA. The short notation of communalities is ' h^2 ' and defined as sum of squared factor loading. In table 12.10, the first column is the name of variables used in our PCA. The second column is the 'initials'. The initial value of the communalities in PCA is 1. The second column 'extraction' shows the proportion of each variable's variation captured by PCA. This value ranges between 0 and 1. When the value is near to 1 that means the variable is well represented and the reverse for value is closer to '0'.

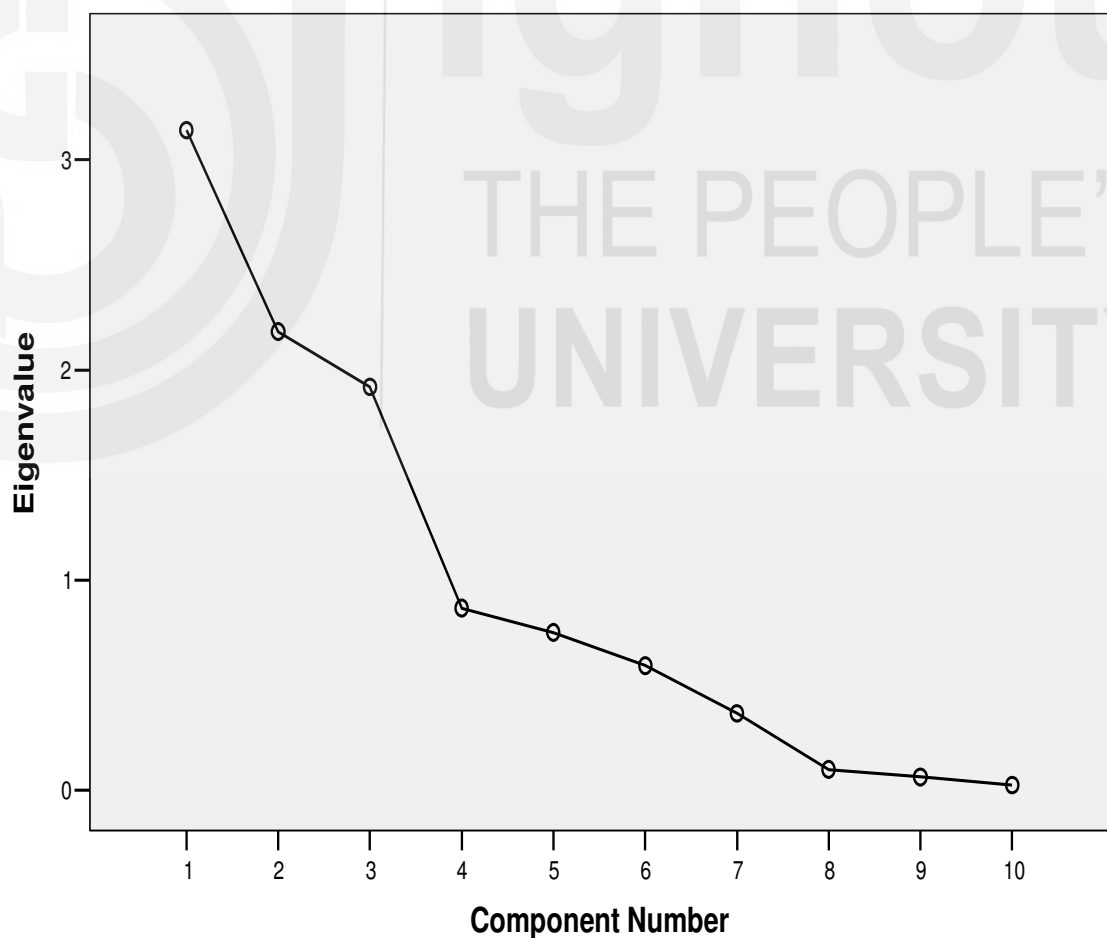
If the communality is low for an item, the reason might be that the item was poorly designed. If the item has very little variance the communality is low. If the different items usually resulting from large positive or negative skewed the communalities is low. We can take an example that if everyone ticks strongly agree, the variation within the variable is low and the communality is low. If the communality is low, we can either remove the item from the analysis to exclude it from any further analyses or we can treat it as a stand alone variable.

Table 12.11: (Output 2) Total Variance Explained: Extraction Method: Principal Component Analysis.

Component	Initial Eigen values			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	3.141	31.413	31.413	3.141	31.413	31.413
2	2.182	21.820	53.233	2.182	21.820	53.233
3	1.919	19.195	72.428	1.919	19.195	72.428
4	.866	8.658	81.086			
5	.750	7.502	88.587			
6	.593	5.927	94.514			
7	.365	3.645	98.160			
8	.098	.976	99.135			
9	.062	.625	99.760			
10	.024	.240	100.000			
	10.0					

The Table 12.11 shows ‘total variance explained’. The first column shows the number of components. The number of components should be equal to number of variables used for PCA. In our example, we have used 10 variables and hence the total number of component is 10. The second part of the table reflect initial Eigen value which consists of three columns i.e. Total, percentage of Variance, Cumulative percentage. The initial Eigen values are the variances of principal components. Here as we use the standardized values of variable, the variance becomes equivalent to 1 and the total variance is equal to the total number of variables i.e. ‘10’. The second column ‘total’ contains the Eigen values. Here it can easily be seen that the Eigen values of subsequent components gradually reduce implying that the successive components add less and less variation. The third column shows the percentage variation on the components and the fourth column explains the cumulative addition of percentage variation of cumulative percentage components. In our example the first component explain 31.4 per cent of total variation and the second component explain 21.8 percent of total variation and so on. The third row of fourth column shows that the first three components explain 72 per cent of total variation.

The second part of the table is ‘Extraction Sums of Squared Loadings’ which consists of three columns – Total, percentage of Variance and Cumulative percentage. This has actually reproduced the three rows of figures of the first part of table reflecting the components whose eigen value is greater than 1.



Graph 12.1: Scree plot

The 'scree plot' is a graph whose 'X axis' represent the component number and 'Y axis' represents 'Eigen values'. In other words scree plot demonstrates the graph of first two column of 'output 2' table. From the graph one can select the number of components to be taken for analysis. It can be clearly seen from the above graph that from the fourth component onward the line becomes flat indicating that when the successive addition is less and less, the line of the graph becomes more and more flatter.

Table 12.12: (Output 4) Component Matrix(a)

	Component ^a		
	1	2	3
Percentage of other than agricultural labour to all labour	.202	.807	.174
Percentage of net irrigated area to total sown area.	-.185	.793	-.293
Per capita value of agricultural output	-.403	.596	.438
Monthly per capita consumption expenditure	.717	.217	-.378
Female Literacy rate	.837	.406	.105
women work force participation rate	-.721	-.233	-.533
Percentage of households having access to safe drinking water	-.504	-.041	.730
Percentage of villages having access to paved road	.733	-.291	.389
Percentage of villages having access to primary health centre	.441	-.417	.451
Average Casual wage rate	-.430	.145	.535

Extraction Method: Principal Component Analysis.

a 3 components extracted.

Table 12.12 (output 4) shows Component Matrix of first three components. The first column is the name of variable used in PCA. The values in the output table are the component loading which are the correlation between the variables and components. The values in the tables range from -1 to +1. One can note that the sum of square of all three component matrix of a variable is equal to the extraction value of the communalities of that variable. In this example three component has been extracted as the eigen value for these component is greater than 1.

Table 12.13: (Output 5) Reproduced Correlations

	Percentage of other than agricultural labour to all labour	Percentage of net irrigated area to total sown area	Per capita value of agricultural output	Per capita consumption expenditure	Female Literacy rate	women work force participation rate	Percentage of households getting safe drinking water	Percentage of villages having access to paved road	Percentage of villages having access to primary health centre	Average casual wage rate
Reproduced Correlation										
Percentage of other than agricultural labour to all labour	0.721	0.554	0.477	0.253	0.512	-0.424	-0.007	-0.022	-0.171	0.120
Percentage of net irrigated to total sown area	0.554	0.748	0.419	0.151	0.137	0.102	-0.152	-0.479	-0.543	0.032
Per capita value of agricultural output	0.477	0.419	0.710	-0.324	-0.049	-0.083	0.500	-0.297	-0.228	0.491
Per capita consumption expenditure	0.253	0.151	-0.324	0.702	0.648	-0.366	-0.645	0.314	0.055	-0.479
Female Literacy rate	0.512	0.137	-0.049	0.648	0.876	-0.754	-0.362	0.536	0.247	-0.246
women work force participation rate	-0.424	0.102	-0.083	-0.366	-0.754	0.858	-0.015	-0.668	-0.460	-0.010
Percentage of households having access to safe drinking water	-0.007	-0.152	0.500	-0.645	-0.362	-0.015	0.789	-0.074	0.123	0.603
Percentage of villages having access to paved road	-0.022	-0.479	-0.297	0.314	0.536	-0.668	-0.074	0.773	0.620	-0.144
Percentage of village access to primary health centre	-0.171	-0.543	-0.228	0.055	0.247	-0.460	0.123	0.620	0.572	-0.005
Average Casual wage rate	0.120	0.032	0.491	-0.479	-0.246	-0.010	0.603	-0.144	-0.005	0.491
Residual(a)										
Percentage of other than agricultural labour to all labour		-0.149	-0.039	-0.161	-0.040	0.002	-0.087	-0.161	0.154	-0.134
Percentage of net irrigated area to total sown area	-0.149		-0.069	0.096	-0.066	-0.030	0.121	0.111	0.016	0.033
Per capita value of agricultural output	-0.039	-0.069		0.139	-0.028	0.145	0.045	0.089	0.046	-0.180
Monthly per capita consumption expenditure	-0.161	0.096	0.139		-0.039	0.119	0.120	0.084	0.067	0.045
Female Literacy rate	-0.040	-0.066	-0.028	-0.039		0.008	-0.082	-0.054	-0.052	0.151
women work force participation rate	0.002	-0.030	0.145	0.119	0.008		0.001	-0.016	0.147	-0.025
Percentage of households having access to safe drinking water	-0.087	0.121	0.045	0.120	-0.082	0.001		0.143	-0.070	-0.173
Percentage of villages having access to paved road	-0.161	0.111	0.089	0.084	-0.054	-0.016	0.143		-0.183	-0.113
Percentage of villages having access to primary health centre	0.154	0.016	0.046	0.067	-0.052	0.147	-0.070	-0.183		-0.007
Average Casual wage rate	-0.134	0.033	-0.180	0.045	0.151	-0.025	-0.173	-0.113	-0.007	

Extraction Method: Principal Component Analysis.

- Residuals are computed between observed and reproduced correlations. There are 29 (64.0%) no redundant residuals with absolute values greater than 0.05.
- Reproduced communalities